

(2.3) Basics of quantum mechanics

Dr. Arun Sehrawat

video

Basic elements of QM: Preparation and Measurement

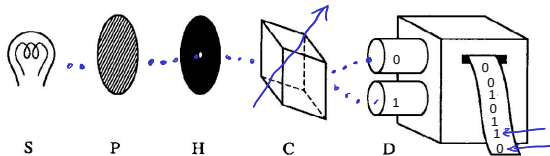
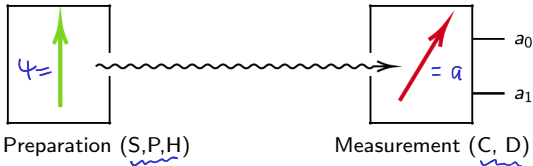


Fig. 1.3. Light from a thermal source S passes through a polarizer P, a pinhole H, a calcite crystal C, and then it triggers one of the detectors D. The latter register their output in a device which prints the results.

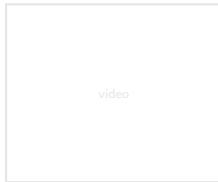
- “P” prepares photons (light) in a particular polarization ψ
- “C” measures polarization in a particular direction $\underline{a}$
- “D” detects photons and prints the data



Data (classical):

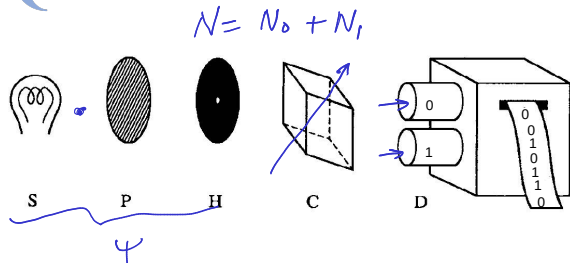
$\nearrow$	$\swarrow$	$\swarrow$	$\nearrow$	$\swarrow$	$\nearrow$	...
a_0	a_1	a_1	a_0	a_1	a_0	...

dir (pointing to a_0) *outcome* (pointing to a_1)



A. Peres, *Quantum Theory: Concepts and Methods* (Kluwer Academic Publisher, 1995), Chapt 1. $\leftarrow$
 R. P. Feynman, *The Feynman Lectures on Physics*, Vol 3, Chaps 1, 5, and 6.
 B.-G. Englert, *Lectures on Quantum Mechanics - Vol 1: Basic Matters* (World Scientific, 2006), Chapt 1 and 2.

A quantum property: Outcomes are "fundamentally" random



• Classical electrodynamics can explain that the light intensity goes to D_0 is $\frac{N_0}{N_0+N_1}$ and D_1 is $\frac{N_1}{N_0+N_1}$, approximately.

• When the next photon (indivisible light particle) come, **no-one** can tell whether it will go to D_0 or D_1 . Hence, the outcomes are fundamentally random. Based the preparation or data, quantum mechanics can only provide their probabilities $p_0 \approx \frac{N_0}{N_0+N_1}$ and $p_1 \approx \frac{N_1}{N_0+N_1}$.

- Photons are independent and identically prepared (iid).
- Outcomes are mutually exclusive.

Data (sequence of outcomes):

Sample {

$\nearrow$	$\swarrow$	$\swarrow$	$\nearrow$	$\swarrow$	$\nearrow$	...
a_0	a_1	a_1	a_0	a_1	a_0	...

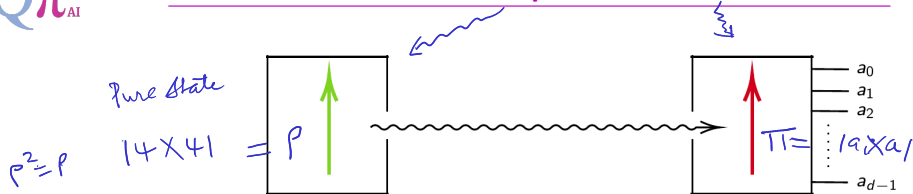
{OR Code
OTP

Applications:

- Quantum random number generator
- Quantum cryptography (BB84 QKD protocol)
- Sampling from a probability distribution

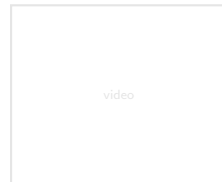
(p_0, p_1)

Basic elements of QM: Preparation and Measurement



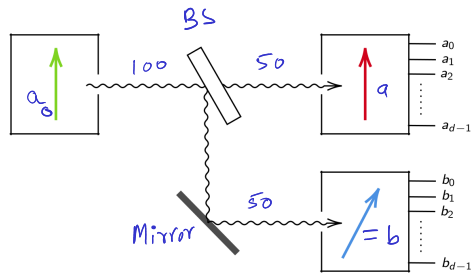
$$p_i = \text{tr}(\rho \Pi_i)$$

- ρ summarizes (what we know about state of the system) the preparation procedure.
Knowledge
- Π_i describes the measurement procedure (apparatus).
- p_i is what we estimate from the observed data (after repeating the same experiment many times).



Distinct measurements

$$\rho = |a_0\rangle\langle a_0|$$



$$B_a = \{|a_i\rangle\}$$

$$\sum_i |a_i\rangle\langle a_i| = I$$

Data: $a_0 \ a_0 \ a_0 \ a_0 \ a_0 \ a_0 \ a_0 \ a_0 \ a_0 \dots$ *fixed*

Probabilities $p_i = |\langle a_i | a_0 \rangle|^2 \in \{0, 1\}$ means outcome is certain (**NO superposition**)

$$B_b = \{|b_j\rangle\}$$

$$\sum_j |b_j\rangle\langle b_j| = I$$

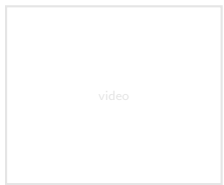
Data: $b_0 \ b_1 \ b_1 \ b_0 \ b_2 \ b_1 \ b_{d-1} \ b_2 \dots$

Probabilities $0 \leq q_j = |\langle b_j | a_0 \rangle|^2 < 1$ means outcomes are uncertain (**superposition**).

$$|a_0\rangle = \sum_j |b_j\rangle \langle b_j | a_0 \rangle$$

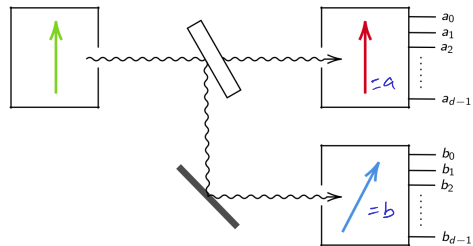
- From the same ρ , one can generate infinitely many distinct probability distributions $\{p_i\}, \{q_j\}, \dots$ by choosing different measurement settings.

- Uncertainty relation: $-\ln(\max_{ij} |\langle a_i | b_j \rangle|^2) \leq \underbrace{-\sum_i p_i \ln(p_i)}_a - \underbrace{\sum_j q_j \ln(q_j)}_b$
Shannon Entropy



Distinct measurements

$$\rho = |a_0\rangle\langle a_0|$$



$$\sum_i |a_i\rangle\langle a_i| = I$$

Data: $a_0 \ a_0 \ a_0 \ a_0 \ a_0 \ a_0 \ a_0 \ a_0 \ \dots$

Probabilities $p_i = |\langle a_i|a_0\rangle|^2 \in \{0, 1\}$ means outcome is certain (**NO superposition**)

$$\sum_j |b_j\rangle\langle b_j| = I$$

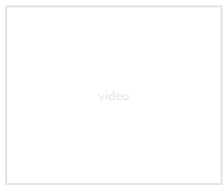
Data: $b_0 \ b_1 \ b_1 \ b_0 \ b_2 \ b_1 \ b_{d-1} \ b_2 \ \dots$

Probabilities $0 \leq q_j = |\langle b_j|a_0\rangle|^2 < 1$ means outcomes are uncertain (**superposition**).

- Unitary transformation $U = \sum_i |b_i\rangle\langle a_i|$ changes the orthonormal bases

$\{|a_i\rangle\} \mapsto \{|b_i\rangle\}$. $U^\dagger$ is its inverse, hence $U^\dagger U = I = UU^\dagger$.

Handwritten notes: $\underbrace{\{|a_i\rangle\}}_{B_a} \mapsto \underbrace{\{|b_i\rangle\}}_{B_b} \quad \hookrightarrow \sum_i |a_i\rangle\langle b_i|$

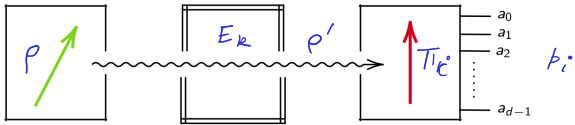


$|ψ\rangle$

CPTP

Proj, Ba

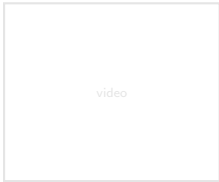
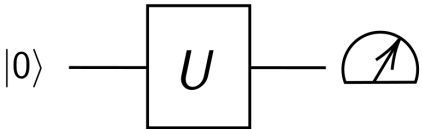
	Quantum state ρ	Evolution $\{E_k\}$	Measurement (POVM) $\{\Pi_i\}$	Probabilities $\{p_i\}$
Positivity:	$0 \leq \rho$	$\sum_k E_k \rho E_k^\dagger$	$0 \leq \Pi_i$	$0 \leq p_i = \text{tr}(\rho \Pi_i)$
Normalization:	$\text{tr}(\rho) = 1$	$\sum_k E_k^\dagger E_k = I$	$\sum_i \Pi_i = I$	$\sum_i p_i = 1$



Preparation

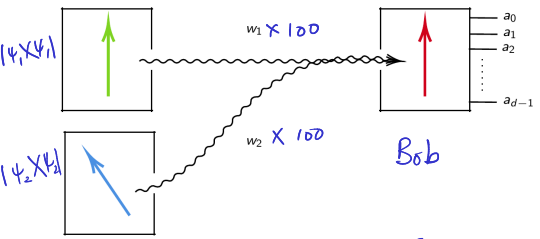
Evolution

Measurement



video

Q π_{AI} Mixed state



Alice

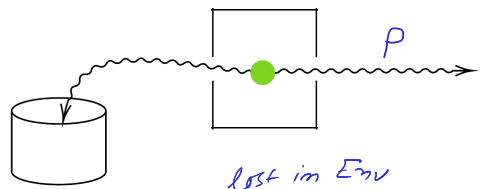
Mixture: convex combination

$$\rho = w_1 |\psi_1\rangle\langle\psi_1| + w_2 |\psi_2\rangle\langle\psi_2|,$$

where $0 \leq w_j \leq 1$ and $\sum_j w_j = 1$

Application: Quantum cryptography (BB84 QKD protocol)

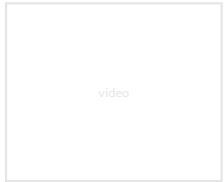
2



lost in Env

- When a composite system is in a pure entangled state and we trace out one of its subsystem, then the rest will end up in a mixed state (decoherence). *open qm. syst.*

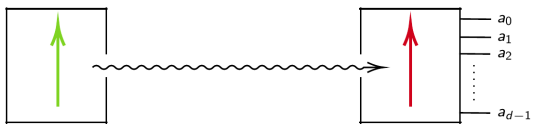
Application: Quantum cryptography (E91 QKD protocol)



C. H. Bennett and G. Brassard, Proceedings of IEEE International Conference on Computers, Systems and Signal Processing, pp. 175-179, (1984).
A. Ekert, Phys. Rev. Lett. **67**, 661 (1991).

Q π_{AI} Projector: mathematical description of measurement procedure

$$\begin{aligned}
 (\langle \psi | \phi \rangle)^+ &= \overline{\langle \psi | \phi \rangle} \\
 &= \langle \phi | \psi \rangle \\
 &= \langle \phi | \psi \rangle
 \end{aligned}$$

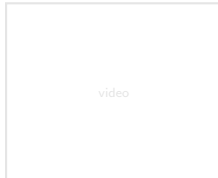


$$B = \{|a_i\rangle\}$$

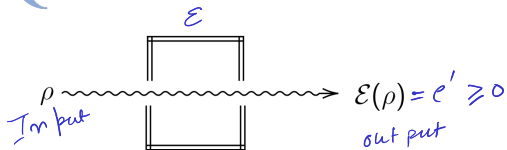
- For each outcome a_i , we associate a projector $\Pi_i = |a_i\rangle\langle a_i| \equiv \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \end{pmatrix} \begin{pmatrix} 0 \cdots 1 \cdots 0 \end{pmatrix} = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$.
 $\Pi = \Pi^+ = \Pi^2$ $1^2 = 1$ $0^2 = 0$

- Their collection $\{\Pi_i\}_{i=0}^{d-1}$ describes a complete projective measurement: $\sum_{i=0}^{d-1} \Pi_i = I = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$

- Born rule:** $p_i = |\langle a_i | \psi \rangle|^2 = \langle \psi | a_i \rangle \langle a_i | \psi \rangle = \langle \psi | \Pi_i | \psi \rangle = \text{tr}(|\psi\rangle\langle\psi| \Pi_i)$
 $\text{tr}(AB) = \text{tr}(BA)$
 $\rho \leftarrow \text{State density of stat op}$
 Inner prod.



Quantum channel: Completely Positive Trace Preserving (CPTP) map

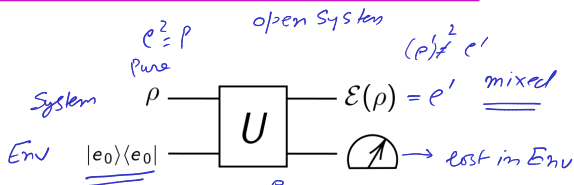


Completely Positive: $(\mathcal{I} \otimes \mathcal{E})(\rho) \geq 0$

Trace Preserving: $\text{tr}(\mathcal{E}(\rho)) = 1$

Transpose is a positive map but not a CP map. If ρ is an entangled state and $\mathcal{E}$ is transpose, then $(\mathcal{I} \otimes \mathcal{E})(\rho) = \rho'$ will not be a positive operator (legitimate state) [**PPT creation for entanglement detection**].

at least one e.v. -ve

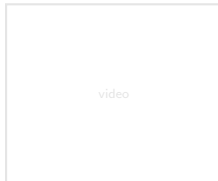


$$\mathcal{E}(\rho) = \sum_k \langle e_k | U (\rho \otimes |e_0\rangle\langle e_0|) U^\dagger | e_k \rangle = \text{tr}_E(\rho_{SE})$$

$$= \sum_k E_k \rho E_k^\dagger, \quad \text{where } E_k = \langle e_k | U | e_0 \rangle.$$

OSR

Entanglement and then partial trace with respect to environment.



M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2000), Chapt 8.

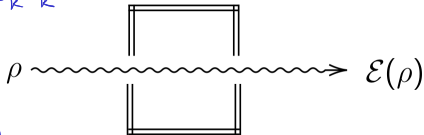
A. Peres, Phys. Rev. Lett. **77**, 1413 (1996).

Quantum channel: Completely Positive Trace Preserving (CPTP) map

$$P_k = |a_k\rangle\langle a_k| = E_k = E_k^\dagger = E_k^\dagger E_k$$

$$\rho_k = \frac{|a_k\rangle\langle a_k| \rho |a_k\rangle\langle a_k|}{\langle a_k | \rho | a_k \rangle}$$

$$= |a_k\rangle\langle a_k|$$



A map is CP iff $\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger$ and $\sum_k E_k^\dagger E_k \leq I$ (equality of TP).

— Unitary evolution: $\mathcal{E}(\rho) = U \rho U^\dagger$ and $U^\dagger U = I = V V^\dagger$

— Measurement: $\mathcal{E}(\rho) = \sum_k p_k \frac{E_k \rho E_k^\dagger}{p_k}$ and $\sum_k \underbrace{E_k^\dagger E_k}_{\Pi_k} = I$ Comp

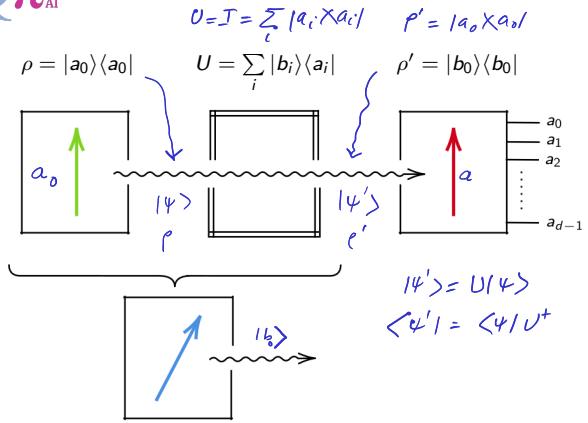
$$\sum_k p_k |a_k\rangle\langle a_k| = \sum_k p_k \rho_k$$

Probability of getting k th outcome: $p_k = \text{tr}(\rho \Pi_k)$

State of system after k th outcome: ρ_k
 $\uparrow$ $\uparrow$
 prep meas

M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2000), Chapt 8.

Quantum evolution and superposition



$$U = I = \sum_i |a_i\rangle\langle a_i| \quad \rho' = |a_0\rangle\langle a_0|$$

$$U = \sum_i |b_i\rangle\langle a_i| \quad \rho' = |b_0\rangle\langle b_0|$$

$$|\psi'\rangle = U|\psi\rangle$$

$$\langle\psi'| = \langle\psi|U^\dagger$$

$$\mathcal{B}_a = \{|a_i\rangle\}$$

Data: $a_0 \ a_0 \ a_0 \ \dots$ Fixed

Not
get
 a_i

Data: $a_0 \ a_0 \ \underline{a_2} \ a_0 \ \underline{a_1} \ \underline{a_1} \ \underline{a_{d-1}} \ \underline{a_{d-1}} \ \dots$ Super position

Evolved state: $|\psi'\rangle\langle\psi'| = U|\psi\rangle\langle\psi|U^\dagger$

$\underbrace{\hspace{1cm}}_{\rho'} \quad \underbrace{\hspace{1cm}}_{\rho}$

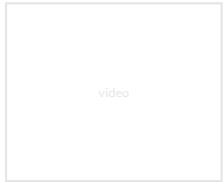
Measurement: $|\psi'\rangle = \sum_i |a_i\rangle\langle a_i|\psi'\rangle$

$\underbrace{\hspace{1cm}}_{|b_0\rangle} \quad \underbrace{\hspace{1cm}}_I \quad \text{amplitude}$

a little bit change of basis, superposition emerges

Probability: $p_i = |\langle a_i|\psi'\rangle|^2$

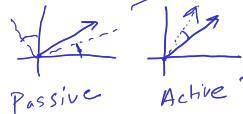
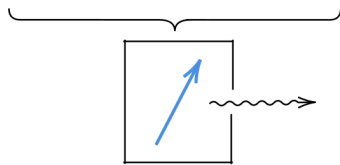
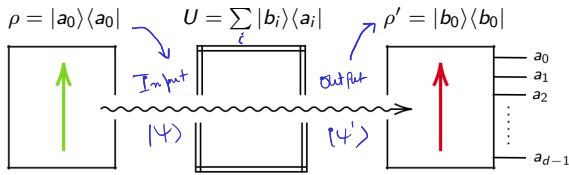
meas $\nearrow$ cros state



Quantum evolution and superposition

$$i = \sqrt{-1}$$

$$U = \exp(-i H t / \hbar)$$



amp $\langle \bar{a}_i | \psi \rangle = \psi(q, t)$
 $\nwarrow$ wave fun

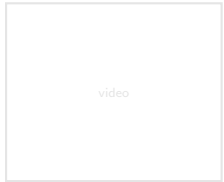
Observable: $A = \sum_i a_i |a_i\rangle \langle a_i| = A^\dagger$
 Av. value $\langle A \rangle = \langle \psi' | A | \psi' \rangle = \sum_i a_i p_i$
 $\text{tr}(e'A)$

measurement
 evolved
 $p_i = |\langle a_i | \psi' \rangle|^2$
 $= \langle a_i | \psi' \rangle \langle \psi' | a_i \rangle$
 $= \langle a_i | \underbrace{U | \psi \rangle \langle \psi | U^\dagger}_{\text{evolution of state}} | a_i \rangle$
 $= \langle \psi' | a_i \rangle \langle a_i | \psi' \rangle$
 $= \langle \psi | \underbrace{U^\dagger | a_i \rangle \langle a_i | U}_{\text{evolution of observable}} | \psi \rangle$
 $|\psi'\rangle = U |\psi\rangle$
 output Input
 $\langle \psi' | = \langle \psi | U^\dagger$
 $\rho' = U \rho U^\dagger$

$$a_i \in \mathbb{R}$$

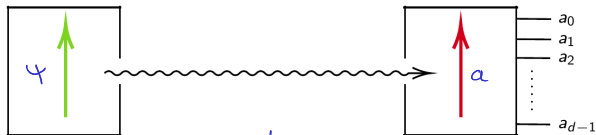
Schrödinger picture: evolution of state $i\hbar \frac{\partial}{\partial t} \langle a_i, t | = \langle a_i, t | H$

Heisenberg picture: evolution of observable $\frac{dA}{dt} = \frac{\partial A}{\partial t} + \frac{1}{i\hbar} (AH - HA)$
 $-i\hbar \frac{\partial}{\partial t} |a, t\rangle = H |a, t\rangle$
 Commutator



A. Peres, *Quantum Theory: Concepts and Methods* (Kluwer Academic Publisher, 1995), Chapt 8.
 B.-G. Englert, *Lectures on Quantum Mechanics - Vol 1: Basic Matters* (World Scientific, 2006), Chapt 3.

Quantum state: mathematical description of preparation procedure



max pos out.

Qudit: a d -level quantum system

- Suppose we have prepared qudit in a state $\psi = \uparrow$. If we measure it in the same direction, then we will always get the same outcome $a_0 = \uparrow$.

Data:

a_0	a_0	a_0	a_0	a_0	a_0	$\dots$
-------	-------	-------	-------	-------	-------	---------

Fix

Pure state vector (ket): $|\psi\rangle = \sum_i |a_i\rangle \langle a_i|\psi\rangle = |a_0\rangle \equiv \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ in $\mathcal{B} = \{|a_i\rangle\}$

$\underbrace{\sum_i}_{I} \underbrace{\langle a_i|\psi\rangle}_{amp}$

$\underbrace{\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{dir}$

$\langle a_0| = (|a_0\rangle)^\dagger \equiv (1 \ 0 \ \dots \ 0)$ is its associated bra.

conj transp

Born rule: probability of getting the a_i outcome

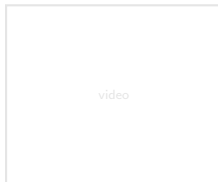
Inner prod

$$p_i = |\langle a_i|\psi\rangle|^2 = \begin{cases} 1 & \text{for } i = 0 \\ 0 & \text{for } i \neq 0 \end{cases}$$

- Normalization $\langle\psi|\psi\rangle = 1$ tells that the sum of probabilities will be 1 for **every** measurement.

$$\langle\psi|\psi\rangle = \sum \langle\psi|a_i\rangle \langle a_i|\psi\rangle = \sum_i P_i = 1$$

- Orthogonality $\langle a_i|a_j\rangle = 0, i \neq j$, means the outcomes are mutually exclusive.



Summary so far

$$\sum_i \pi_i = I$$

Every meas has an outcome

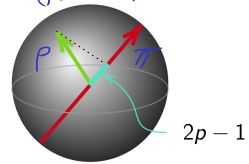
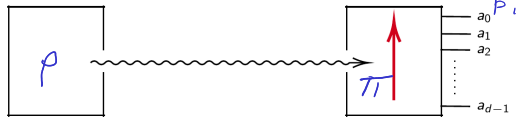
Born Rule:
 $\rho^2 = \rho$

$$p_i = \text{tr}(\rho \pi_i)$$

$$\pi_i^2 = \pi_i$$

	Quantum state ρ	Measurement (POVM) $\{\pi_i\}$	Probabilities $\{p_i\}$
Positivity:	$0 \leq \rho$ ✓	$0 \leq \pi_i$ ✓	$0 \leq p_i$
Normalization:	$\text{tr}(\rho) = 1$ ✓	$\sum_i \pi_i = I$ ✓	$\sum_i p_i = 1$

$$\sum_i \text{tr}(\pi_i \rho) = \text{tr}(\rho) = 1$$



Single qubit
2-level Sys.

